

Chemistry — Assignment 10: Carbocation, Carbanion & Free Radical

Anshuman Atrey · 150096724029 · BTech CSE 2024-28

Chemistry (Foundational Chemistry) · Semester 5

Question. Differentiate carbocation, carbanion and free radical; explain stability; differentiate homolytic/heterolytic fission; explain a carbonium-ion reaction.

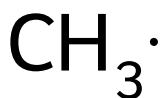
	Carbocation	Carbanion	Free radical
charge	+1	-1	0
valence e ⁻ on C	6	8	7
hybridisation	sp ²	sp ³	sp ²
geometry	trigonal planar	pyramidal	~planar
role	electrophile	nucleophile	highly reactive



methyl cation



methyl anion



methyl radical

Carbocation stability: $3^\circ > 2^\circ > 1^\circ > \text{methyl}$ (+I of alkyl groups, hyperconjugation, resonance e.g. allyl/benzyl).

Carbonium-ion reactions: react with a nucleophile; lose a β -H to give an alkene; or rearrange by a 1,2-hydride/alkyl shift to a more stable carbocation.